BROADBAND MULTIMEDIA SYSTEMS ARCHITECTURE AND INTEGRATION*

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ABSTRACT

Interactive multimedia systems with guaranteed service quality require complex integration of servers, broadband networks and terminals. A critical issue is development of such systems based on standards in order to achieve interoperability comparable to the Internet. The task for devising standardized interactive networked multimedia systems has been undertaken by the Digital Audio Visual Council (DAVIC). In this paper we describe development of a DAVIC compliant networked multimedia system. This system is based on the implementation and integration of hardware and software based on standards developed by DAVIC, MPEG-2 DSM-CC, MHEG-5, CORBA and ATM.

1 INTRODUCTION

Digital Audio Visual Council (DAVIC) has recently defined a complete specification of an end-to-end networked multimedia system for consumer applications. Among the potential uses of such system are Movies on Demand, Teleshopping, Games, Distance learning, Home banking, Internet access, etc. The DAVIC system differs fundamentally from the current Internet by aiming for the provision of guaranteed high quality of service. This in turn implies that the DAVIC system requires high quality digital broadband network with specific delivery parameters. Such a network is not available today for consumers but progress in technology and new services can make it viable soon. By creating a system integrating multimedia computing and broadband communication technology DAVIC responds to these future needs. Presently, implementation of the DAVIC system is a nontrivial task since it requires integration of many novel technologies and substantial software development. In this paper we review the DAVIC system and underlying standards and point out the aspects of its practical realization.

2 DAVIC SYSTEM

DAVIC sysem is intended for network delivery of high-quality content to home users. Networking system is based on the ATM (Asynchronous Transfer Mode) network because of its connectivity and resource management. ATM connectivity enables to establish links for reliable sending of streams with specified bit rate. As shown in the DAVIC connectivity scenario at Fig. 1, Each STU (Set Top Unit) can connect to any server in the ATM network with guaranteed quality of service. Such system needs controlling available network resources, marked in Fig. 1 as the Session and Resource Manager (SRM).

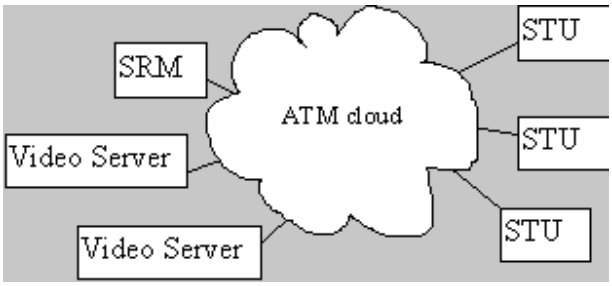


Figure 1: DAVIC system scenario

3 DAVIC COMMUNICATION

In the ATM networking systems there is a provision for flexible connections establishment based on numbering of terminal devices and bandwidth allocation. This is called SVC (Switched Virtual Connection) and underlying protocol is called Q.2931. DAVIC information flow is based on the Q.2931 ATM [4,5] signalling protocol. The broadband data streaming channel in the DAVIC system is the same MPEG-2 TS (Transport Stream) stream as used in the digital television standards DVB (Digital Video Broadcasting) in Europe, but it is send over an ATM virtual circuit established between the server and the STU using the Q.2931 protocol. Additional channel is used for control flows which in the DAVIC system are provided by the Digital Storage Media Command and Control (DSM CC) protocol which is part of the ISO MPEG-2 standard. DSM CC plays critical role in the stream control by the STU.

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3.1 DSM CC Functional Reference Model

The Digital Storage Media Command and Control (DSM CC) [3] specification is a set of protocols intended to provide the control functions and operations specific for managing MPEG-2 video streams in heterogeneous network environments including applications like video-on-demand and interactive video. There are two kind of messages in it. DSM CC User to Network (U-N) messages set up and release sessions. The SRM entity controls the session establishment and allocation of system resources ensuring quality of service. DSM CC User to User (U-U) messages provide communication between applications running on the client side and on the server side. The DSM CC User to User protocol provides communication between the STU and the server using CORBA (Common Object Request Broker Architecture) distributed computing system. To open a session the STU sends a U-N Session Request message to the SRM. The SRM informs the server and waits for its acknowledgment. Once the session is accepted by the server, the SRM informs the STU. At this point the STU can directly access the server applications with the U-U protocol. For Video On Demand Application, the U-U protocol allows the STU to select a film and control its delivery over S1 with VCR-like functions (Pause Stream, Resume Stream, Fast Forward or Fast Backward).

3.2 CORBA

CORBA is a standard for building platform independent distributed computer systems developed by the OMG (Object Management Group) [7]. CORBA provides services for distributed applications including object naming, security, transactions and persistence services. CORBA specifies a client/server architecture that is location independent and language neutral. CORBA provides special Interface Definition Language (IDL) which has maps on standard computer languages, communication model and mechanism with protocols for achieving full interoperability of applications running on different platforms. An ORB (Object Request Broker) [7] is a special architectural model for connecting objects requesting services to the objects providing them. It provides transport mechanisms which sends calls from the client to the target object. The ORB activates object servers when a client request needs to be served. The model is shown in Fig. 2. Messages are send via the ORB which effectively is a kind of software bus common for all system platforms implementing it.

3.2.1 CORBA Object Reference

Modern architecture of CORBA, its universality and its platform independence are primary factors why it is adopted by the DSM-CC as its system solution. In DAVIC, the STU uses object references to remotely browse and access application objects residing in servers, and activate specific operations like e. g. in the case of video streams Pause Stream, Resume Stream, Fast Forward or Fast Backward. An STU uses the CORBA resolve operation to obtain a reference to an object instance from a name. Another important aspect of CORBA is special protocol called IIOP (Internet Inter ORB Protocol) by which mechanism is provided for many different ORB implementations to communicate with one another (Fig. 3). IIOP is a protocol running directly on top of the standard TCP/IP protocol of the Internet and it does not require the use of higher level RPC (Request Procedural Call) mechanism used in standard computers systems.

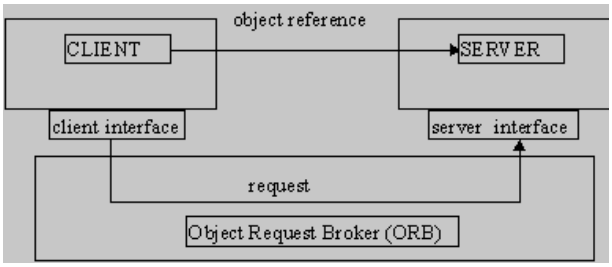


Figure 2: ORB Object Request Broker

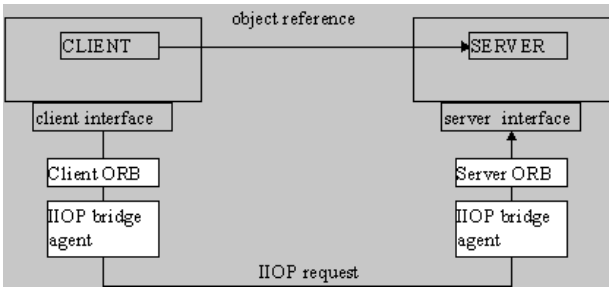


Figure 3: IIOP

IIOP defines the Common Data Representation (CDR) which is a data transfer syntax mapping between different ORBs. IIOP defines messages exchanged between agents to locate objects implementation and manage communication channels. IIOP specifies how agents open connections and use them to transfer IIOP messages over TCP/IP. In the DSM CC, IIOP is the transport layer for CORBA remote procedure calls. In the chosen scenario, TCP/IP packets carrying IIOP protocol over which DSM-CC messages are sent using special ATM SVC link. The MPEG TS broadband flow uses a different ATM SVC. Figure 4 summarizes the roles of the different entities and the related information flow.

The STU, SRM and server communicate through U-N messages to establish and release sessions. Once a session is established, an ATM SVC connection is opened between the server and the STU. Over this channel, the STU can issue CORBA requests for applications, e.g. specific films residing on a server. The server then opens another ATM SVC to the STU through which the MPEG-2 TS stream is sent by the video pump in the server to the MPEG-2 decoder in the STU. The STU

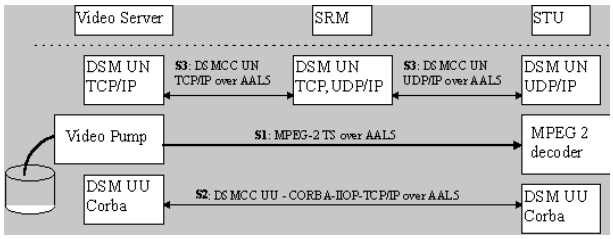


Figure 4: DAVIC information flow

can then display images of the film. Specific CORBA requests can inform the server video pump to stop the MPEG-2 stream or fast forward/backward the film. The DSM CC does not only define VCR type of command but the the STU can also make SQL queries on the server's databases, read/write files, or load MHEG or JAVA scripts.

4 IMPLEMENTATION OF DAVIC COMPLIANT SERVER SYSTEM

DAVIC compliant media server requires development and integration of several major components. A video pump is necessary for generation of multiple media streams and it needs to be tightly integrated with the ATM communication links. The DSM CC server is responsible for handling U-N and U-U commands of the DSM CC and controlling the video streams. The hardware necessary for the media server has been integrated and all the necessary software was developed by us within the EU ACTS projects DAM and IMMP. Our implementation of DAVIC compliant media server runs on SUN server with Solaris 2.x UNIX operating system. Solaris has real time process management features which are used for guaranteeing processing priorities for handling video streams. We use Sparc Storage Arrays as mass storage for video streams and SUN ATM adapter providing 155 Mbits/s and the Q.2931 protocol. The DSM CC User-to-User commands between the client and server are sent over a CORBA (RPC) Remote Procedure Call mechanism. The CORBA compliant ORB provided in SUN NEO package supports the Internet Inter-ORB Protocol (IIOP) for client-server signalling. The implementation of the DSM CC U-U server agent is done using the Interface Definition Language (IDL) definition taken from the DSM CC standard. The IDL objects interface is analyzed and compiled into the C++ skeleton code by a specific compiler provided by the SUN NEO software. The skeleton source code is a running empty server. Each interface function defined in the IDL files is mapped into a C++ class method. From the skeleton one can implement the functionality specified in the DSM CC, for example File read, File Write, Stream play, Stream stop. Each DSM CC object is mapped into a class object. The instantiation of such a class creates in the SUN ORB naming space the related DSM CC object. When a user request is sent to

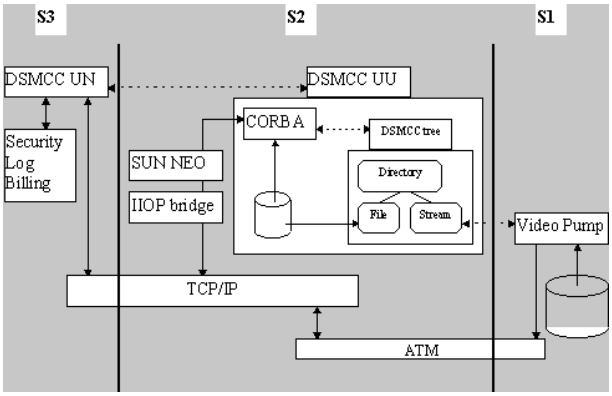


Figure 5: Server Architecture

the ORB for a created DSM CC object, the ORB executes the corresponding C++ class method. There is only one copy of the tree of DSM CC objects representing DSMC-CC applications. All users share the same objects on the server. When called, each server object knows the identity of the calling client by checking the subobject identifier provided by SUN NEO. The subobject identifier technique allows a server object to have different addresses. Each address refers to a particular caller. A subobject identifier consists of two parts. The first is the ORB address which points in the ORB address space and allows the ORB to find the requested object. The ORB address is the same in every subobject identifier pointing to the same object. The second part however specifies the calling user. In our implementation it is a number allocated when the session is opened. This number is just passed through by the ORB to the C++ class method. The class method is able to identify the calling user with the number passed as a parameter to the method.

Figure 5 shows the general server architecture. The DSM CC U-N module deals with incoming requests for new sessions from the SRM on behalf of STUs. This module runs the DSM CC U-N protocol as defined by DAVIC and DSM CC. In relation with the DSM CC U-U module, it provides an authentication mechanism, checks users right to access objects, logs the traffic and has basic tools for billing. The IIOP bridge gets incoming IIOP CORBA requests from an STU and translates them into NEO style for the SUN NEO ORB. In return, the response from the NEO ORB is translated into an IIOP packet and sent back to the STU. The DSM CC U-U module uses the NEO ORB. It handles requests from STU and activates the suitable applications. For Video On Demand, the VCR requests are forwarded to the Video Pump module which controls the MPEG-2 stream flow. The other applications available are file access, games and the loading of MHEG and JAVA scripts. The Video Pump module controls MPEG-2 flows to the different STUs. It controls the sending rate and the quality of service. Under the control of the Video On

Demand application, the Video Pump can pause or resume the MPEG-2 stream to a particular STU.

4.1 Implementation of DSM CC CORBA objects

DSM CC defines CORBA objects accessible remotely by the STU. Through those objects, the STU activates applications on the server. The three main type of objects are Stream, File and Directory. Each type of object has a associated set of commands. Stream is the base object for Video On Demand applications. Associated with a film, e. g. MPEG file, it allows the STU to remotely execute VCR type of commands. The main Stream commands are: Pause, Resume, Jump and Play (for direction and speed). With the File object an STU can read or write files stored on the server. Multimedia access is largely read-only. The STU retrieves MHEG [1] or Java scripts from the server using File objects. The Directory object builds a tree of objects. The STU can remotely browse the tree and select a requested object. The browsing can be done under human control or under the control of an application downloaded into the STU. When opening a session, the client gets from the server an object reference of a Directory object. Starting from this object, the user can browse the DSM CC tree and select specific application.

5 INTEGRATION OF CLIENT

The DAVIC DSM CC client mirrors in many respects the server (Figure 6). Basic media type is MPEG-2 TS but for applications with spatiotemporally synchronized data standardized format according to the MHEG standard is used. On the client side MHEG engine implementation for application playback is needed. Other substantial hardware and software integration is also required in order to interface ATM network and playback of applications. We have integrated a PC based client in cooperation with partners from ACTS project DAM. For MPEG-2 playback a hardware card from Optibase is used. This card is used with the display adapter card Flashpoint from Integral Technologies. ATM interface is realized using 25 Mb/s link card from Virata. Using software development kits for the cards it was possible to integrate them to play back in real time applications involving MPEG-2 streams of up to 8 Mb/s coming from the media server through the ATM card. DSM CC client is implemented using Orbix CORBA client for PC running under the MHEG engine.

One of the main problems in the complete DAVIC system implementation is ensuring interoperability by strict adherence to the standard and debugging the system. This is nontrivial problem because all system components are new and not tested yet exhaustively in practice. Many problems had to be solved before the system was ready to operation. This includes hardware for MPEG-2 playback from the network and CORBA used for the DSM-CC.

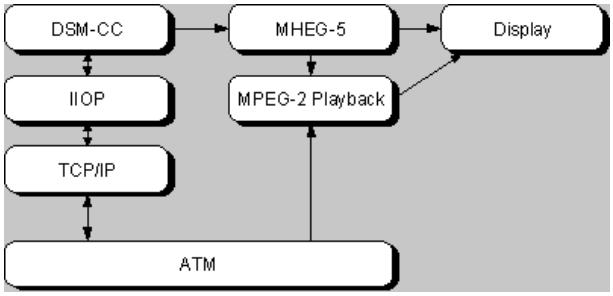


Figure 6: Client architecture

6 CONCLUSION

We have implemented and tested an interactive networked multimedia system with guaranteed quality of service. The system is based on the DAVIC standard and requires integration of many hardware and software components which need to be tightly linked to achieve correct operation. Implementation of the highly quality multimedia system according to the DAVIC specifications has been a nontrivial task requiring multidisciplinary cooperation and extensive infrastructure for testbed development and testing.

7 ACKNOWLEDGEMENTS

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